

UTS Cloud Computing Class - Lab Guide

Week 4 Lab: vCenter Server and VM migration

This week the lab tasks will involve configuring and setting up a vCenter Server, and connecting your two ESXi hosts to it. Once connections are established, we can begin performing live VM migrations which are a foundation technology to our virtual datacenter.

Task 1. Prepare for the lab.

1. Open VMware Workstation, and **power on** the following five VMs: 'ESXi01', 'ESXi02', and 'iSCSI'.
2. From the Mozilla Firefox browser, connect to the servers ESXi01 and ESXi02 using the URL. If not working, use the same method as last week to fix the network connections.
* **Hint:** Steps under 'To configure IP address', Week 2
3. Verify that your hosts are connected to the iSCSI target from last week.
* **Hint:** Check under 'Storage' for each ESXi
4. Verify that there is already a vm on each host from earlier weeks.

** STOP * If you have no iSCSI connectivity either follow the steps outlined from last week.*

Task 2. Configure and connect to vCenter Server (VCSA).

1. **Power On** OpenWRT VM on your workstation.
2. **Power On** VCSA VM, and open a browser tab for it.
3. From your web browser browse to your VCSA VM's **Management** portal: <https://vcsa.vsphere.local:5480>

You will see an error "Exception in invoking authentication handler User password expired". This occurs because the password needs to be changed every three months.

To update the password: (1). Open a terminal (Applications->Utilities->Terminal)

(2) input the following command: `ssh root@172.20.20.94`

Input: yes

Input password: Vmware1!

Input command: shell

Input command: passwd

Follow the prompts to set a new password.

4. Login with the root account, and the password '**VMware1!**'.
5. Once completed, you should be able to login to the vcsa machine using the **Administration** portal: <https://vcsa.vsphere.local:443> or using the URL <https://172.20.20.94>. The browser will give an option as '**LAUNCH VSPHERE CLIENT (HTML5)**'. Click on it to go to the Login window.
6. Login with the credentials as:

Username: **administrator@vsphere.local** Password:
VMware1!

Task 3. Verify Licensing for vCenter and hosts.

Currently, you should have a web page open to your vcsa on port 443 or IP address 172.20.20.94. From this Administrative portal, we can manage vCenter, and all the hosts and vm's connected to it.

The required licenses have been installed in the provided files. No further action is required for this task. Below are the instructions for installing and managing the licenses, should you need them in the future.

1. **Manage licensing.** On the home page of vCenter, at the top of the window, there is a warning message about expired or expiring licenses. Click on the **'MANAGE YOUR LICENSES'** button and the page to enter new licenses will open. You are required to obtain a new license keys before proceeding any further.
2. **New License Keys.**
You are required to get two licenses: one for hosts and other for the vCenter server. From the above link you must add two VMware products to the cart.
 - **VMware vCenter Server 7.x Standard**
 - **VMware vSphere 7.x Enterprise Plus**
3. **Add new licenses.** On the vCenter browser Licenses page, click on **'Add New Licenses'** to include the vCenter and vSphere Enterprise Plus licenses.
4. Next go to **'Assets'** tab, select the **'VCENTER SERVER SYSTEMS'** asset and click the option above it **'Assign License'** to assign the new vCenter license and click **'OK'**.
5. Select the next option of **'HOSTS'**, select the two hosts and again **'Assign License'**. Select the vSphere license and click **'OK'**.
6. Once you have assigned the licenses, refresh the browser to see if the licenses are verified and that the License Expiration warning message is gone.

Task 4. Migrate virtual machines to shared storage.

Migrate a powered on VM to shared storage. This task is called a 'Storage vMotion' where we can move a virtual machines files between two different locations without having to shut the virtual machine down.

1. **Power On** the VM that is on ESXi1 2. Right-Click the VM and select **Migrate**.
3. Select **'Change storage only'**, and choose to move the VM across to shared storage you created earlier **'YourName-vmfs1'**

4. To confirm the VM have migrated to shared storage, click on the **VM summary page -> VM Hardware section -> Hard disk 1 -> Location**. This property should show the location of the VM i.e. '**YourName-vmfs1**'.

Task 5. Migrate virtual machines to another host.

With the Virtual Machine on shared storage, it should be possible to perform a vMotion of the running state to another host.

1. Select your running VM that is on shared storage.
2. Right-Click the VM and select **Migrate**.
3. This time select '**Change compute resource only**' and choose to move the VM across to the other host. (for example, if your VM is on ESXi01, move it to ESXi02.

Hints:

- If you get an error in '**Compatibility**' section after selecting another host, this might be because vMotion is not enabled. Please refer to next page "How is vMotion enabled?"
- After selecting the Host you want the VM to be changed to, you might see the error "Currently connected device 'CD/DVD drive 1' uses backing '[Datastore1] TinyCore-current.iso', which is not accessible." Why? How to fix it?

Task 6. Configure vMotion on both hosts.

To perform VM migration between two hosts, vMotion property should be enabled on both the hosts.

1. Select ESXi1 (172.20.20.51). Navigate to '**Configure**' tab, select '**Networking->Virtual Switches**' option.
2. Under '**Management Network**', select '...' in front of '**vmk0 : 172.20.20.51**' and select '**Edit Settings**'.
3. Check the box for '**vMotion**' and click '**OK**'.
4. Perform steps 1-3 for ESXi2 to enable '**vMotion**' for ESXi2.

Task 7. Prepare for the next lab.

Once you have verified connectivity to shared storage, and have successfully performed a storage vMotion, and a host vMotion you can shut down your virtual machines in preparation for next week's lab.